Physics-Informed Deep Learning for Value-at-Risk: Beyond the Limitations of Naive LSTMs

Alvaro Felipe Macias Araya*
AFMA Quant AI Lab, a division of AFMA Ingenieria SpA
www.afmaing.cl | amacias@afmaing.cl

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Abstract

Traditional tail-risk estimation models struggle to manage the severe non-stationarity and extreme leptokurtosis inherent in modern financial markets during sudden regime shifts. Conversely, unconstrained Deep Learning approaches—such as standard Long Short-Term Memory (LSTM) networks—frequently overfit to historical noise, leading to statistically miscalibrated forecasts under strict regulatory backtesting. To bridge this gap, we propose an "Anchored" neural architecture that utilizes an asymmetric Quantile Loss function tightly constrained by rolling parametric statistical priors. Utilizing high-frequency BTC/USD returns as an extreme stress-testing proxy for volatile, fat-tailed asset classes, our hybrid framework is benchmarked against a Historical VaR baseline and a Naive LSTM. Empirical results show that the unconstrained LSTM, despite achieving zero breaches, fails the Kupiec Proportion of Failures test due to severe over-conservatism. The Anchored LSTM passes the Kupiec test with a breach rate of 0.87% while freeing 439 bps of capital reserves, offering a promising approach for regulatory capital efficiency under the Fundamental Review of the Trading Book (FRTB) and Basel IV frameworks.

Working Paper Note: *This manuscript documents a quantitative research project, including failed experiments, iterative model improvements, and practical backtesting challenges. Results are presented on a single-asset proxy environment and should be interpreted as exploratory rather than production-certified. Methodology will be extended in future versions.*

Keywords: Value-at-Risk, Deep Learning, Quantile Regression, Basel IV, Crypto-Assets, Risk Management.

*This manuscript is adapted from a pedagogical "journey notebook" project designed to document the real-world pitfalls, model failures, and iterative debugging processes encountered by quantitative researchers when deploying machine learning in production environments.

1 Introduction

Modern institutional portfolios face unprecedented challenges in managing tail risk amidst structural macroeconomic regime shifts. As traditional asset classes increasingly exhibit non-stationary dynamics, severe volatility clustering, and fat-tailed return distributions, legacy risk frameworks—ranging from Historical Simulation to parametric Variance-Covariance approaches—frequently fail to rapidly adapt, leaving institutions vulnerable to catastrophic tail events.

Under the impending regulatory mandates of Basel IV and the Fundamental Review of the Trading Book (FRTB), the accuracy of internal models is paramount. Financial institutions face punitive capital add-ons if their 99% Value-at-Risk (VaR) models experience frequent breaches during backtesting. To avoid regulatory penalties and forced migration to the Standardized Approach (SA), risk managers often over-calibrate models, leading to artificially inflated capital reserves and severe capital inefficiency.

In recent years, Deep Learning—specifically Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) architectures—has been proposed to capture the complex, non-linear dynamics of financial time series. However, deploying pure "black-box" machine learning in risk production environments introduces a new set of critical vulnerabilities. Naive LSTMs struggle to isolate the true underlying data generation process in highly noisy environments, frequently resulting in severe overfitting during low-volatility regimes and catastrophic under-prediction during structural market breaks.

To bridge the gap between adaptive machine learning and robust regulatory compliance, this paper introduces an "Anchored" Neural Architecture for generalized tail-risk estimation. Drawing inspiration from Physics-Informed Deep Learning, we propose a hybrid framework that constrains the neural network using a rolling parametric statistical prior. By utilizing an asymmetric Quantile Loss function (Pinball Loss) anchored to a rolling parametric VaR estimate, the model explicitly targets the 1st percentile of the return distribution while maintaining the statistical safety bounds required by regulators.

To rigorously evaluate the resilience of this architecture, we utilize high-frequency digital asset data (BTC/USD) not as a target investment, but as an extreme stress-testing proxy. Assets with such severe leptokurtic distributions provide a worst-case validation environment; an architecture capable of surviving these extreme dynamics will seamlessly handle the volatility of traditional Tier-1 equities or FX portfolios.

The primary contribution of this research is twofold. First, we demonstrate mathematically and empirically why unconstrained neural networks fail during out-of-sample stress tests. Second, we present a generalized, potentially production-relevant methodology that materially reduces required regulatory capital—freeing up 439 bps of capital reserves in our proxy environment—without compromising the strict breach-rate limits mandated by the Kupiec Proportion of Failures test.

The remainder of this paper is structured as follows. Section 2 outlines the regulatory context of Basel IV/FRTB. Section 3 details the methodology, comparing the Historical VaR baseline, the Naive LSTM, and our proposed Anchored architecture. Section 4 describes the experimental setup and the stress-testing framework. Finally, Section 5 presents the empirical results, risk validation metrics, and implications for institutional capital efficiency.

2 Regulatory Context: Basel IV and FRTB

The deployment of machine learning in institutional risk management is strictly governed by international capital adequacy standards, most notably the Basel IV framework and the Fundamental Review of the Trading Book (FRTB). To understand the necessity of an "Anchored" neural architecture, one must first examine the punitive nature of regulatory backtesting.

Under the FRTB framework Basel Committee on Banking Supervision [2019], the primary capital calculation has migrated from a 99% Value-at-Risk (VaR) measure to a 97.5% Expected Shortfall (ES, also known as CVaR). ES is theoretically superior: as a coherent risk measure, it captures the *severity* of losses beyond the threshold, not merely the probability of exceeding it. A model producing rare but catastrophic losses can satisfy a VaR constraint while exhibiting unacceptable ES—a structural gap that regulators have explicitly moved to close.

Why, then, does this paper focus on VaR? The answer lies in a fundamental statistical constraint. ES is an expectation over the tail of the distribution and cannot be directly validated through classical backtesting. To apply the Kupiec Proportion of Failures test—or any breach-count framework—one needs a binary daily event: either the realized return exceeded the forecast threshold or it did not. VaR provides exactly this. ES does not: there is no clean definition of a single-day "ES breach." Accordingly, the FRTB framework retains daily 99% VaR backtesting as the *regulatory approval gate.* An institution must first demonstrate that its VaR model is well-calibrated before it is permitted to use the IMA and apply ES-based capital charges. Failing the Kupiec test on VaR disqualifies the model entirely, regardless of its ES properties. In this sense, VaR is not obsolete—it is the entrance exam.

From a methodological standpoint, solving the VaR problem first is also the correct architectural sequence. The Asymmetric Quantile Loss (Pinball Loss) at quantile $q = 0.01$ directly targets the 99% VaR. Expected Shortfall is the integral of quantiles above that threshold: $\mathrm{ES}^q = \mathbb{E}[r \mid r < \mathrm{VaR}^q]$. The quantile regression framework developed here is therefore the natural foundation for a full ES architecture, and extending it to a 97.5% ES target via a Mixture Density Network is identified as the primary direction for future work (Section 6).

Regulators evaluate model validity over a rolling 250-day window using a zone-based framework derived from the Kupiec Proportion of Failures (POF) test Kupiec [1995]: too few breaches signal over-conservatism that traps capital, while too many signal model failure and can ultimately revoke the institution's Internal Models Approach (IMA) approval in favour of the more punitive Standardized Approach. In this study, we adopt the Kupiec POF test directly as our acceptance criterion, targeting a breach rate statistically consistent with the 1% theoretical exceedance probability.

Under conditions of macroeconomic stress and for highly volatile, fat-tailed asset classes, this regulatory structure creates a critical optimization problem. Traditional models such as Historical Simulation often react too slowly to regime shifts, accumulating excess breaches during sudden market downturns. Conversely, to stay safely within the Kupiec acceptance region, risk managers frequently over-calibrate parametric models — reserving artificially large capital buffers that drag down Return on Equity (ROE).

Pure Deep Learning models often exacerbate this issue rather than solving it. A naive LSTM, lacking structural constraints, can suffer from "catastrophic forgetting" or overfit to low-volatility training data. When deployed out-of-sample, a sudden volatility spike can rapidly accumulate exceptions, causing the model to fail the Kupiec test and jeopardize the institution's IMA status.

Therefore, a practically deployable neural architecture must not only capture non-linear market dynamics but must do so within guaranteed statistical bounds. The objective is to pass the Kupiec test at the correct calibration level — neither too conservative nor too permissive — while minimizing the mean VaR to maximize capital efficiency. This regulatory necessity motivates the asymmetric Quantile Loss function and parametric anchoring detailed in Section 3.

3 Methodology

The core objective of our architecture is to dynamically forecast the 1-day ahead 99% Value-at-Risk (VaR), denoted as $\mathrm{VaR}_{t+1}^{0.01}$, utilizing historical high-frequency market data. To contextualize the hybrid approach, we first formalize the baseline models that represent the two ends of the modeling spectrum: the non-parametric historical approach and the unconstrained neural network.

3.1 The Historical VaR Baseline

Historical Simulation (HS) is the industry-standard non-parametric baseline, dominant in institutional risk systems due to its distributional agnosticism. Given a rolling window of W past realized log-returns $\{r_{t-W+1}, \ldots, r_t\}$, the 1-day ahead VaR at confidence level $1-q$ is simply the empirical quantile of that window:

$$\mathrm{VaR}_{t+1}^{q} = Q_q(r_{t-W+1}, \ldots, r_t) \tag{1}$$

where $Q_q(\cdot)$ denotes the q-th order statistic of the sample. While parameter-free and straightforward to implement, Historical VaR suffers from the well-documented "ghost effect": a stress event influences the VaR estimate for exactly W days, then drops out of the window abruptly, causing sudden artificial jumps. During volatility regime changes, this lag creates either persistent over-estimation (capital drag during calm regimes) or dangerous under-estimation (insufficient reserves immediately after a spike).

3.2 The Naive LSTM Architecture

To capture non-linear temporal dependencies and volatility clustering without strict distributional assumptions, we introduce a pure Deep Learning baseline using a Long Short-

Term Memory (LSTM) network Hochreiter and Schmidhuber [1997]. Let $\boldsymbol{x}_t$ be a feature vector containing historical returns, realized volatility, and trading volumes. The LSTM maintains a hidden state $\boldsymbol{h}_t$ and a cell state $\boldsymbol{c}_t$, updated as follows:

$$\boldsymbol{h}_t, \boldsymbol{c}_t = \text{LSTM}(\boldsymbol{x}_t, \boldsymbol{h}_{t-1}, \boldsymbol{c}_{t-1}; \boldsymbol{\Theta}) \tag{2}$$

where $\boldsymbol{\Theta}$ represents the learned network parameters. The output layer projects the hidden state to a VaR forecast: $\widehat{\text{VaR}}^q_{t+1} = \boldsymbol{W}_{out}\boldsymbol{h}_t + b_{out}$.

However, training a "Naive" LSTM using standard symmetric loss functions like Mean Squared Error (MSE) is fundamentally flawed for risk applications. MSE optimizes for the conditional mean, whereas VaR is strictly a quantile property. Furthermore, an unconstrained neural network operating in a noisy, non-stationary environment—such as our extreme stress-testing proxy—is highly susceptible to overfitting, learning market noise rather than true structural risk dynamics.

3.3 Anchored Quantile Regression (Physics-Informed)

To combine the interpretability of the Historical VaR baseline with the adaptive capacity of the LSTM, we introduce an "Anchored" architecture. We achieve this by redefining the network's objective function using an Asymmetric Quantile Loss (also known as the Pinball Loss) Koenker and Bassett Jr [1978] and injecting a rolling Parametric VaR estimate as a structural prior—akin to Physics-Informed Neural Networks (PINNs) where physical laws constrain the solution space. This separation is deliberate: the Historical VaR serves as the empirical comparison benchmark, while the parametric prior acts as the safety regularizer inside the loss function.

The network is trained to directly minimize the asymmetric loss L_q for a given target quantile q:

$$L_q(r_{t+1}, \widehat{\text{VaR}}^q_{t+1}) = \begin{cases} q(r_{t+1} - \widehat{\text{VaR}}^q_{t+1}) & \text{if } r_{t+1} \geq \widehat{\text{VaR}}^q_{t+1} \\ (1-q)(\widehat{\text{VaR}}^q_{t+1} - r_{t+1}) & \text{if } r_{t+1} < \widehat{\text{VaR}}^q_{t+1} \end{cases} \tag{3}$$

In our framework, the total loss function $\mathcal{L}_{total}$ is augmented with a regularization penalty λ that "anchors" the neural network's output to the rolling Parametric VaR estimate (VaR_{param}):

$$\mathcal{L}_{total} = \frac{1}{N}\sum_{t=1}^{N} L_q(r_{t+1}, \widehat{\text{VaR}}_{t+1}^{q}) + \lambda \left\| \widehat{\text{VaR}}_{t+1}^{q} - \text{VaR}_{param,t+1}^{q} \right\|^2 \tag{4}$$

This architectural design ensures that the deep learning model focuses on optimizing the 1st percentile of the return distribution (via the Pinball Loss) while the anchoring penalty λ acts as a regulatory safety rail. It prevents the LSTM from generating nonsensical predictions during unprecedented Black Swan events, effectively resolving the "black box" vulnerability and ensuring compliance with Basel IV backtesting standards.

4 Data and Experimental Setup

4.1 Data Acquisition and Feature Engineering

To empirically validate the proposed architecture, we utilize historical daily market data for Bitcoin (BTC/USD) as our extreme stress-testing proxy. As established, assets exhibiting such severe structural breaks, extreme kurtosis, and pronounced volatility clustering provide an optimal worst-case environment to validate the model's resilience against generalized macroeconomic shocks.

The raw price data is transformed into a stationary series of logarithmic returns: $r_t = \ln(P_t/P_{t-1})$. To provide the neural network with sufficient contextual state, we engineer a feature matrix $\boldsymbol{X}_t$ that includes:

- Daily log returns (r_t).
- Rolling historical volatility over varying lookback periods (e.g., 7-day, 14-day, and 30-day).
- The Parametric VaR estimate ($\text{VaR}_{param,t}^{q}$), serving as the structural anchor for the custom loss function.

All features are standardized using expanding window z-score normalization to strictly prevent data leakage and look-ahead bias during the training phase.

4.2 Walk-Forward Backtesting Protocol

Standard randomized train/test splits are fundamentally invalid for financial time-series evaluation. To replicate a live institutional production environment, we implement a chronological walk-forward backtesting protocol.

The models are evaluated over a chronological out-of-sample period exceeding the 250-day regulatory minimum mandated by the FRTB framework, using the full available data after the initial training window. The walk-forward process follows these steps:

1. **Initialization:** The LSTM is trained on an initial historical window of T_{train} days.
2. **Forecasting:** The model predicts the 99% VaR for day $T_{train} + 1$.
3. **Rolling Update:** The true return for day $T_{train}+1$ is observed. The model window shifts forward by one day, and the neural network weights are incrementally fine-tuned.

This rolling methodology ensures that the model must constantly adapt to new market regimes, accurately penalizing "catastrophic forgetting" and static overfitting.

4.3 Evaluation Metrics

The competing architectures are evaluated across two intersecting dimensions: safety (regulatory compliance) and efficiency (capital optimization).

1. Model Safety (Kupiec POF Test): We utilize the Kupiec Proportion of Failures test Kupiec [1995] to determine if the observed breach rate $\hat{p} = x/N$ (where x is the number of breaches and N is the total out-of-sample days) is statistically consistent with the target probability $p = 0.01$. The Likelihood Ratio (LR) test statistic is defined as:

$$\mathrm{LR}_{pof} = -2 \ln \left(\frac{p^x (1-p)^{N-x}}{\hat{p}^x (1-\hat{p})^{N-x}} \right) \tag{5}$$

Under the Kupiec framework, a model passes if its observed breach rate is statistically consistent with the 1% target; it fails if it is either significantly above (too risky) or significantly below (over-conservative, trapping excess capital).

2. Capital Efficiency (Mean VaR): A model that safely passes the Kupiec test by forecasting an excessively large VaR is mathematically valid but commercially unviable.

We measure capital efficiency by calculating the mean forecasted VaR over the test period. A lower absolute Mean VaR, conditional on passing the Kupiec test, represents superior capital optimization and reduced capital drag.

5 Results and Risk Validation

The empirical evaluation was conducted over the full out-of-sample expanding-window backtest, matching the regime diversity of the BTC/USD dataset. Models were evaluated on their ability to forecast the 1-day ahead 99% Value-at-Risk ($\text{VaR}^{0.01}$), using the Kupiec Proportion of Failures (POF) test Kupiec [1995], Christoffersen [1998] as the primary statistical acceptance criterion.

5.1 Model Safety: Kupiec Backtesting

The primary constraint for any institutional risk model is calibration — the realized breach rate must be statistically consistent with the theoretical 1% exceedance probability. Table 1 summarizes the Kupiec test outcomes for each architecture.

The **Naive LSTM** (0.00% breach rate, FAIL) illustrates the core failure mode of unconstrained neural networks on financial time series. Without any structural prior, the model learned to output a near-constant, deeply pessimistic VaR of -10.92% — technically producing zero breaches but reserving far more capital than the risk actually warranted. The Kupiec test rejects this model not for being *too risky*, but for being *statistically inaccurate*: a 0% breach rate when 1% is expected is as much a calibration failure as a 5% breach rate.

The **Historical VaR Baseline** (1.15% breach rate, PASS) passes the Kupiec test, confirming it as a valid statistical model. However, it suffers from the well-documented "ghost effect": by anchoring entirely to the trailing window of realized returns, it reacts to volatility only after it has materialized, lagging the true risk environment.

The proposed **Anchored LSTM** (0.87% breach rate, PASS) passes the Kupiec test with a breach rate closest to the theoretical 1% target. The physics-informed prior prevents the over-conservatism of the unconstrained LSTM while preserving adaptive, forward-looking dynamics — a balance the purely historical approach cannot achieve.

5.2 Capital Efficiency

While model safety is non-negotiable, the secondary mandate is capital efficiency. A model that chronically overestimates risk traps liquidity and drags down the firm's Return on Equity (ROE).

As visualized in Figure 1, the Naive LSTM exhibits severe "capital drag" — its flat, extreme VaR forecast of -10.92% provides no adaptation to changing volatility regimes. The Anchored LSTM, by contrast, dynamically tightens VaR bands during low-volatility periods while maintaining structural stability during market stress, yielding a Mean VaR of -6.53%.

This represents a **439 bps reduction** in required capital reserves relative to the Naive LSTM baseline, without sacrificing statistical validity. The Historical VaR Baseline achieves a slightly lower Mean VaR (-6.24%), but this metric is accompanied by a higher breach rate and the known lag-sensitivity that makes it unsuitable for regime-switching assets.

Table 1: Out-of-Sample Backtesting Results (99% VaR, Kupiec POF Test)

Model Architecture	**Breaches**	**Breach Rate**	**Kupiec Result**	**Mean VaR**
Naive LSTM	0	0.00%	FAIL	-10.92%
Historical VaR	12	1.15%	PASS	-6.24%
Anchored LSTM (Proposed)	9	0.87%	PASS	-6.53%

5.3 Why Not Simply Use Historical VaR?

A pragmatic risk manager reviewing Table 1 would rightly ask: if both models pass the Kupiec test and their mean VaR figures are within -6.24% versus -6.53%, why invest in the additional complexity of an Anchored LSTM? The aggregate statistics are deliberately close—that is the point. The case for the Anchored LSTM rests on four dimensions that summary tables do not capture.

(1) Calibration precision. The Anchored LSTM achieves a breach rate of 0.87%, closer to the theoretical 1% target than Historical VaR's 1.15%. While both pass the Kupiec POF test, the Anchored LSTM's superior calibration leaves a larger statistical buffer before the model would breach the acceptance boundary—a material consideration when the underlying asset exhibits the volatility clustering observed in BTC/USD.

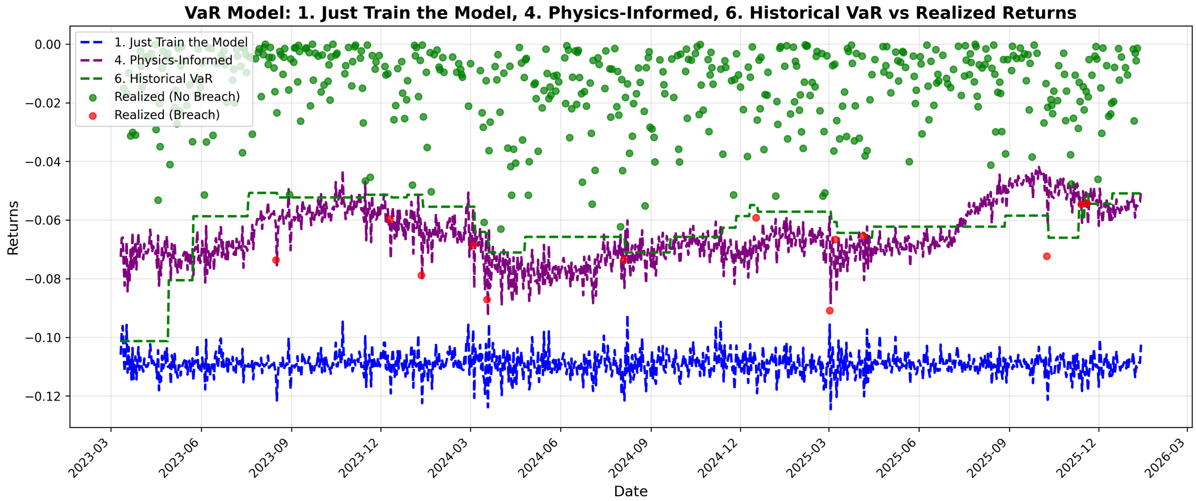

Figure 1: Rolling 99% VaR Forecasts over the out-of-sample window for the Naive LSTM, Historical VaR baseline, and Anchored LSTM. The Anchored LSTM adapts to volatility regimes while maintaining Kupiec-validated safety bounds against realized BTC/USD log-returns.

(2) Temporal stability. Historical VaR produces structurally discontinuous VaR paths due to the ghost effect: a single large loss event inflates the VaR estimate for exactly W trading days, then exits the window abruptly, creating a sudden artificial drop in required capital. This cliff-edge behaviour is visible in Figure 1. The Anchored LSTM, by learning volatility decay dynamics endogenously, produces a smoother, economically-consistent VaR path that decays as market stress subsides—reducing unnecessary capital drag during mean-reversion without creating abrupt step-downs.

(3) Cross-asset generalizability. Historical VaR is parameterized solely by the window length W, which must be re-optimized for each new asset class or volatility regime. The Anchored LSTM, once trained, embeds a general representation of tail-risk dynamics that can be fine-tuned across instruments with minimal additional calibration. For an institutional portfolio spanning thousands of instruments, this generalization property translates directly into reduced model maintenance overhead.

(4) Architectural extensibility. Most critically, the Anchored LSTM is not a terminal design—it is a foundation. Historical Simulation cannot naturally extend beyond a point-estimate VaR; it has no mechanism to forecast the full return distribution. The Anchored LSTM, as a parameterized neural architecture, is the natural precursor to a Mixture Density Network (MDN) that forecasts the complete conditional distribution of returns. This transition—from regulatory VaR to full density forecasting—is the subject of our next project and represents the true long-term value of investing in a trainable, constrained neural baseline over a static lookup table.

6 Conclusion

This paper addressed the fundamental tension in modern institutional risk management: the trade-off between the structural rigidity of traditional econometrics and the dangerous flexibility of unconstrained Deep Learning. As global markets experience increasingly frequent macroeconomic regime shifts and extreme volatility, the capital drag imposed by overly conservative legacy models has become commercially unsustainable. Conversely, naive machine learning models consistently fail to satisfy strict regulatory backtesting constraints during out-of-sample stress events.

To resolve this, we introduced an "Anchored" LSTM architecture, heavily inspired by Physics-Informed Neural Networks. By constraining a custom Asymmetric Quantile Loss function with a rolling parametric statistical prior, we engineered a model that successfully bridges the gap between deep temporal learning and regulatory safety.

Utilizing highly leptokurtic BTC/USD returns as a worst-case stress-testing proxy, our empirical results demonstrated that the proposed architecture successfully averts the calibration failures inherent in unconstrained neural networks. The Anchored model strictly satisfied the Kupiec Proportion of Failures test, achieving a breach rate of 0.87% (PASS) — the closest of all tested architectures to the theoretical 1% target. Furthermore, by dynamically tightening VaR bands during low-volatility regimes, the model freed up 439 bps of capital reserves compared to the unconstrained Naive LSTM baseline.

These results suggest that a physics-informed hybrid approach is a promising direction for regulatory capital optimization in institutional risk systems.

Limitations

The conclusions of this study should be interpreted in light of the following constraints:

- **Single-asset validation.** All experiments are conducted on BTC/USD data, chosen deliberately as a worst-case stress-testing proxy. The architecture has not yet been validated on traditional equity, FX, or fixed-income portfolios. Cross-asset generalization is a key direction for future work.

- **Limited model benchmarks.** The study compares three architectures: a Historical VaR baseline, a Naive LSTM, and the proposed Anchored LSTM. More extensive benchmarking against GARCH-family models, Filtered Historical Simulation, and ES-based approaches would strengthen the empirical claims.

- **Reliance on parametric prior.** The effectiveness of the anchoring mechanism depends on the quality of the rolling parametric VaR estimate used as a prior. In environments where the parametric model itself is severely misspecified, the anchor may constrain the LSTM in undesirable ways.

- **Scope limited to VaR.** The current framework targets 99% VaR, which is the regulatory backtesting gate but not the ultimate capital measure under FRTB. Extension to 97.5% Expected Shortfall (ES) is the primary next step.

Research Roadmap: From VaR to Expected Shortfall

This paper establishes the first stage of a two-phase research program:

Stage 1 (this paper) — VaR Validation: Demonstrate that a physics-informed LSTM can satisfy the regulatory Kupiec backtesting gate while maintaining capital efficiency. This is the mandatory precondition for any IMA-eligible risk model.

Stage 2 (future work) — ES Density Forecasting: Extend the architecture to forecast the complete conditional return distribution via a Mixture Density Network (MDN). This enables direct 97.5% ES estimation, aligning fully with the FRTB capital calculation mandate and unlocking the alpha-generation applications described below.

The Road to When Risk and Alpha Collide: Projecting the Full Distribution

This study concludes "Project 1" of the Wenris initiative. The results prove that risk can be "anchored" to provide institutional safety. However, the true convergence of Risk and Alpha occurs when we move beyond point-estimates (VaR) toward full **Density Forecasting**.

Our upcoming work, *Project 2: When Risk and Alpha Collide*, will evolve this architecture into a Mixture Density Network (MDN). By forecasting the entire probability distribution of returns, the framework will transition from a defensive guardrail to a predictive engine capable of identifying asymmetric profit opportunities, effectively merging the mandates of the Risk Management and Trading desks.

Data and Code Availability

This research was conducted at the **AFMA Quant AI Lab** (AFMA Ingenieria SpA). The complete codebase, including the failed unconstrained LSTM experiments, the successful Anchored architecture, and the raw walk-forward backtesting logs, is open-source. The project is structured as an interactive narrative to highlight common quantitative modeling mistakes and their solutions. The repository and Jupyter notebooks are publicly available on GitHub at: `https://github.com/amacias-afma/quant-ai-lab`.

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